Aditya Madhawan
24010100 23

Topic _____ Date _____

DBMS Assignment - 1

Section - A

Q1) DBMS:

Ans1) A DBMS is a software that manages data in an organised way using table. It helps user to store, update and retrieve data easily.
It is required to manage large data safely avoid duplication and provides secure access.

Q2) Why DBMS is better than file system

ans2) It stores data in a structured format and allows multiples ways to access it safely.
File system may create duplicate data and are harder to manage.

Q3) Data Abstraction: It means hiding technical detail & showing only reqd. information to user.
- Physical (how data is stored)
- Logical (what data is stored)
- View (what user sees)

Q4) What is Three schema Architecture?

ans) It divides database into 3 layers
- Internal
- Conceptual
- External level

Topic _____ Date _____

It helps in data independence & easy modification

Q5) Data Independence means changing data set structure at one level without affecting other levels. It helps in flexibility and system maintainace.

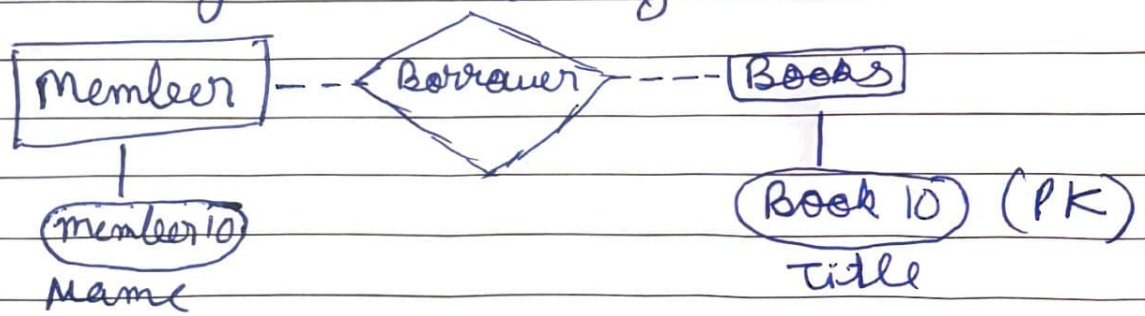
Q6) ER model - It is a diagram that represents entites their attributes and relationships. It is used for data base design planning

Q7) Integrity constrains are rules that ensure correct and valid data.
Ex → Primary keys, foreign keys, Not Null

Section - B

1- Three schema
External level → User view
↓
Conceptual level → Overall structure
↓
Internal level → Physical storage

2. ER diagram → Library



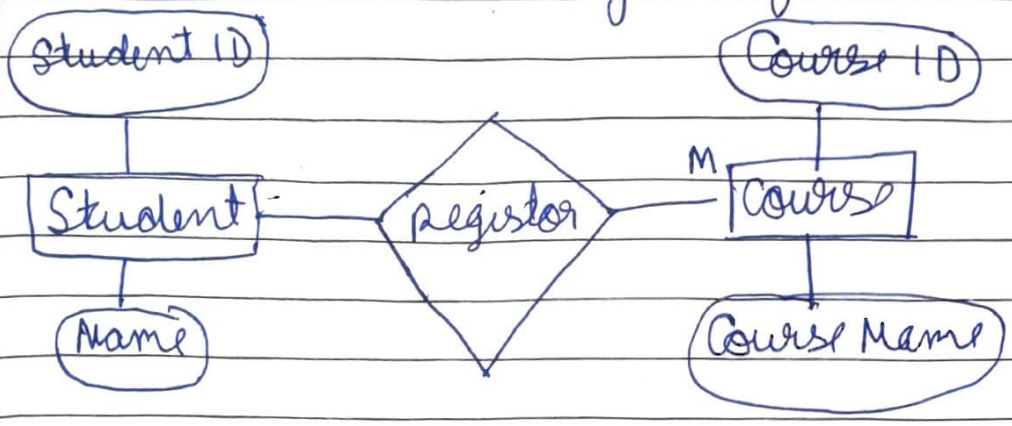
Topic _____ Date _____

- Entities: member, Book
- Relationship - Borrower
- PK - member ID, Book ID

One member can borrow many books

(iii) Student course Registration E R
- Student (student ID, name)
- Course (course ID, ~~name~~ course_name)

Relationship: Student regestor for course



Section - C

(i) ER Diagram - Hospital System
Entities - Patient, Doctor, Appointment
Patient books Appointment
Doctor attends Appointment
Primary keys - Patient ID, Doctor_ID, Appointment ID

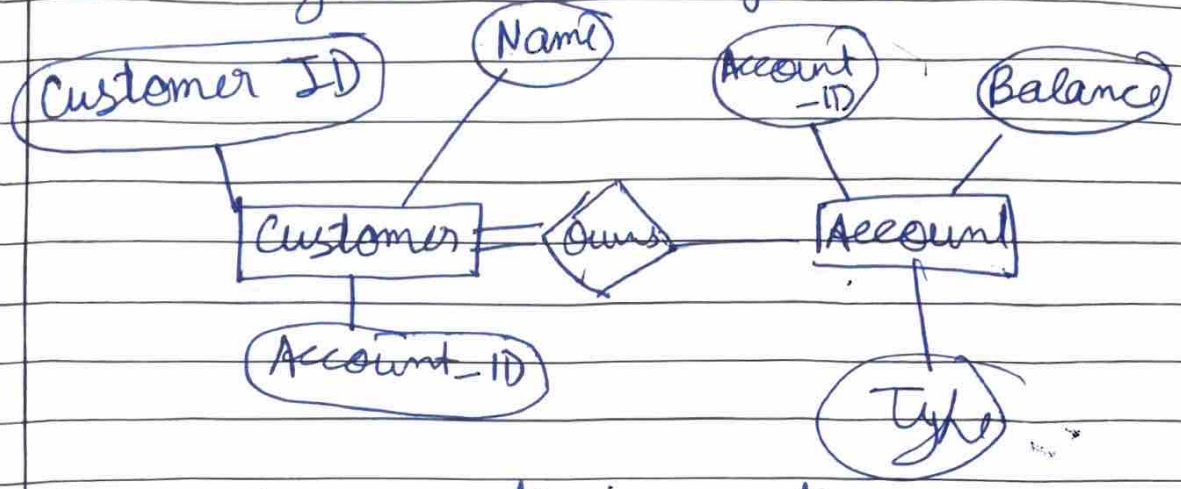
(ii) Online shopping
Customer (Customer ID_PK)
Product (Product_ID_PK)
order (order_ID_PK)
foreign key in order table connect customer & product

Topic _____ Date _____

(iii) University - Data abstraction
student see marks (view level)
Admin manages records (logical level)
DBA. Controls storage (Physical level)

Section - D

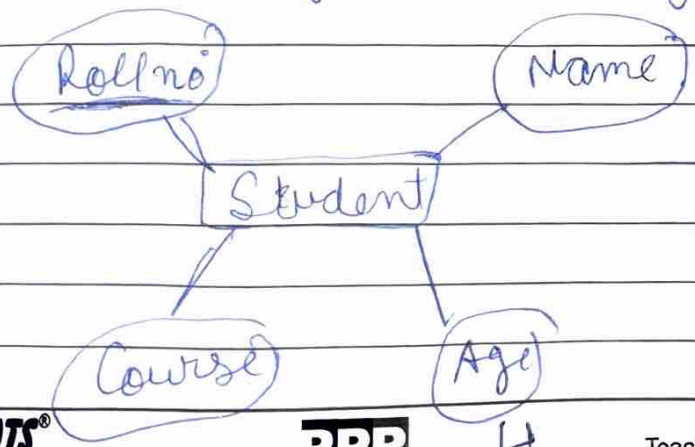
(i) ER diagram - ATM system



Account_no is Primary key
Relation : Customer owns Account
Account_ID : take uniquely. Identifies account

~~Section I~~

Q Draw an entity with attributes and identify its primary key.



Topic _____ Date _____

Entity

→ student

Attributes

Roll no°

Name

course

Age

Primary key = Roll no°

→ Uniquely identifies all records

→ Cannot be null

→ No two students have the same roll no°